BUSFIN 8200: Problem Set 2

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Question 1

(a)

We have complete markets iff we can find a unique matrix Ω s.t. $X\Omega = I$, where

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}, \qquad \Omega = \begin{bmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{bmatrix},$$

and each w_i is a portfolio weight vector using A, B and C. For example, $(10, 20, 30)'$ is asset A's payoff across the three states.

$X\Omega = I$ means

$$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad Xw_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad Xw_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Because X is 3×3, $X\Omega = I$ has a unique solution iff X is invertible, and

$$X \text{ invertible} \iff \det(X) \neq 0 \iff \operatorname{rank}(X) = 3.$$

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}.$$

Row reduction:

$$\begin{aligned} \text{row } 3 - \text{row } 2: &\quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{exchange rows 1 and 2}: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 2 - \text{row } 1: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 3 - 3 \cdot \text{row } 1 - \text{row } 2: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Thus $\operatorname{rank}(X) = 3$ and X is invertible. $\Omega = X^{-1}$ exists and is unique, with $X\Omega = XX^{-1} = I$. Thus, the market is complete.

(b)

Let X be the matrix of $N = 3$ non-redundant asset payoffs, let P be the price vector of the original assets, and let $P_c = (q_1, q_2, q_3)'$ be the price vector of the contingent claims. Use the LOP to solve for P_c as a function of X and P:

$$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}, \qquad X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}.$$

The LOP says that two assets with identical payoffs in every state of nature must have the same price. Therefore,

$$X'P_c = P \quad \Longrightarrow \quad \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Dividing everything by 10:

$$\begin{cases} q_1 + 2q_2 + 3q_3 = 1.6 \\ q_1 + q_2 + q_3 = 0.8 \\ q_2 + q_3 = 0.6 \end{cases} \quad \Longrightarrow \quad \begin{cases} q_1 = 0.2 \\ q_2 = 0.4 \\ q_3 = 0.2 \end{cases}$$

Thus, $P_c = (0.2, 0.4, 0.2)'$.

(c)

$$P_D = \text{Payoff}(D) \cdot P_c = \begin{pmatrix} 30 & 20 & 10 \end{pmatrix} \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 6 + 8 + 2 = 16.$$

(d)

$$P(X) = E[MX] = \sum_{s=1}^{S} \pi(s) M(s) X(s).$$

Since $q_s = \pi(s)M(s)$, we have $M(s) = q_s/\pi(s)$. With

$$P_c = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix} \quad \Longrightarrow \quad M(s) = \begin{pmatrix} 0.8 \\ 0.8 \\ 0.8 \end{pmatrix}.$$

By definition, no arbitrage means that

1. An asset with payoff ≥ 0 must have price ≥ 0.
2. An asset with payoff ≥ 0 and a chance for payoff > 0 must have price > 0.

We have $X \geq 0$, $\Pr(X > 0) > 0$, and $P(X) = E[MX] > 0$ (since $M > 0$). Thus, there is no arbitrage.

Moreover, $P(X) = E[MX] = 0.8\,E[X]$. The SDF stays constant across the three states, which implies it is risk-neutral.

(e)

$$\begin{aligned} P_D = E[MX_D] &= \sum_{s=1}^{S} \pi(s) M(s) X_D(s) \\ &= \tfrac{1}{4} \cdot 0.8 \cdot 30 + \tfrac{1}{2} \cdot 0.8 \cdot 20 + \tfrac{1}{4} \cdot 0.8 \cdot 10 \\ &= 6 + 8 + 2 = 16. \end{aligned}$$

This is the same as in Q1(c).

Question 2

(a)

I construct the unconditional linear SDF as

$$\hat{M}_t^* = \hat{a} - \hat{b}' x R_t,$$

where $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$ and $\hat{a} = 1 + \hat{b}'\hat{\mu}$, imposing the normalization $E[M^*] = 1$.

Table 1 reports the estimated SDF coefficients. The estimated SDF satisfies the unconditional pricing conditions, with $\widehat{E}[\hat{M}^*] = 1$ and $\widehat{E}[\hat{M}^* x R_j] \approx 0$ for all 10 portfolios.

Figure 1 shows no clear monotonic relationship between the estimated b_j coefficients and book-to-market deciles. Instead, the coefficients fluctuate considerably across portfolios, with both positive and negative values.

This pattern arises because the coefficients are determined by both the expected excess returns and the covariance structure of the portfolios, as reflected in $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$. Therefore, portfolios with higher average excess returns do not necessarily receive larger positive coefficients.

Table 1: Estimated BM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to BM decile j; estimation uses decimal monthly excess returns.

$\hat{a}$	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$	$\hat{b}_4$	$\hat{b}_5$	$\hat{b}_6$	$\hat{b}_7$	$\hat{b}_8$	$\hat{b}_9$	$\hat{b}_{10}$
1.0368	-1.5850	3.5503	0.7104	-3.1122	1.1837	2.9972	-5.2931	2.4722	2.8775	-1.1771

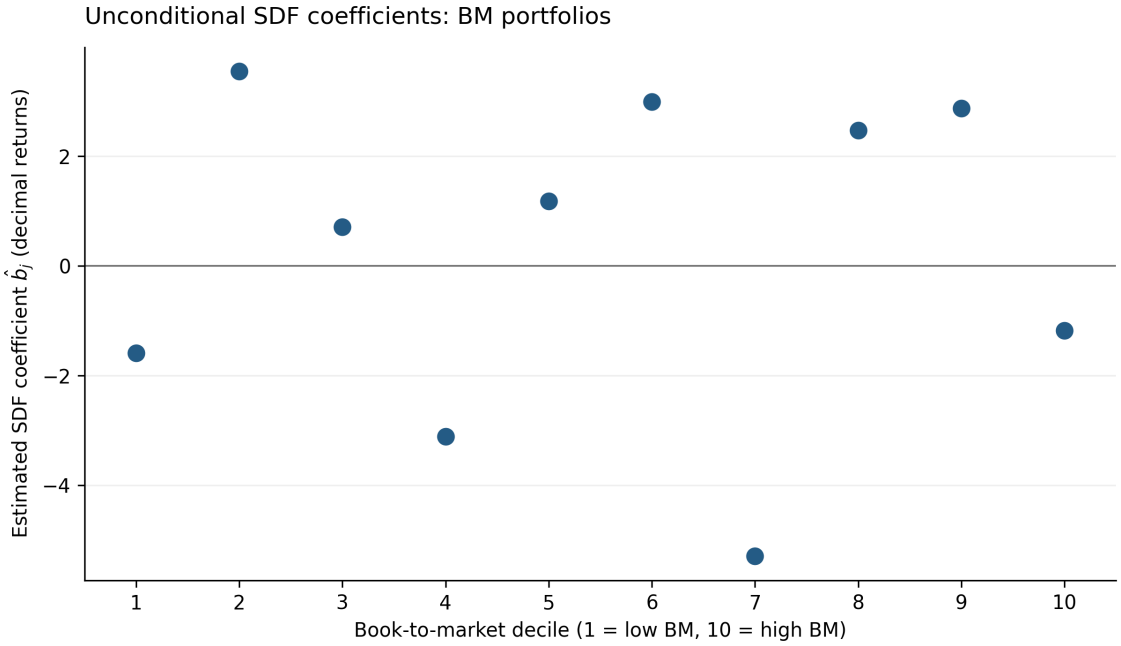


Figure 1: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor T.

(b)

Following the same approach as in part (a), I construct the unconditional linear SDF using the 10 momentum portfolios. Table 2 reports the estimated SDF coefficients.

Figure 2 shows that the estimated coefficients vary considerably across momentum deciles, with both positive and negative values. There is no clear monotonic pattern. As in part (a), this pattern reflects both the expected excess returns and the covariance structure of the portfolios.

The sample correlation between the BM and MOM SDFs is 0.3555, indicating a moderate positive relationship. This suggests that the two SDFs capture some common variation, but also reflect different components of asset returns. The sample correlation between the BM and MOM SDFs is

$$\widehat{\mathrm{Corr}}\left(\hat{M}^{*,BM}, \hat{M}^{*,MOM}\right) = 0.3555.$$

This uses 1,190 aligned monthly observations from July 1927 through August 2026.

Table 2: Estimated MOM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to MOM decile j; estimation uses decimal monthly excess returns.

$\hat{a}$	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$	$\hat{b}_4$	$\hat{b}_5$	$\hat{b}_6$	$\hat{b}_7$	$\hat{b}_8$	$\hat{b}_9$	$\hat{b}_{10}$
1.0666	-4.0194	1.1429	0.1024	1.4897	2.5317	-0.8195	0.2777	0.9818	-0.9414	3.4370

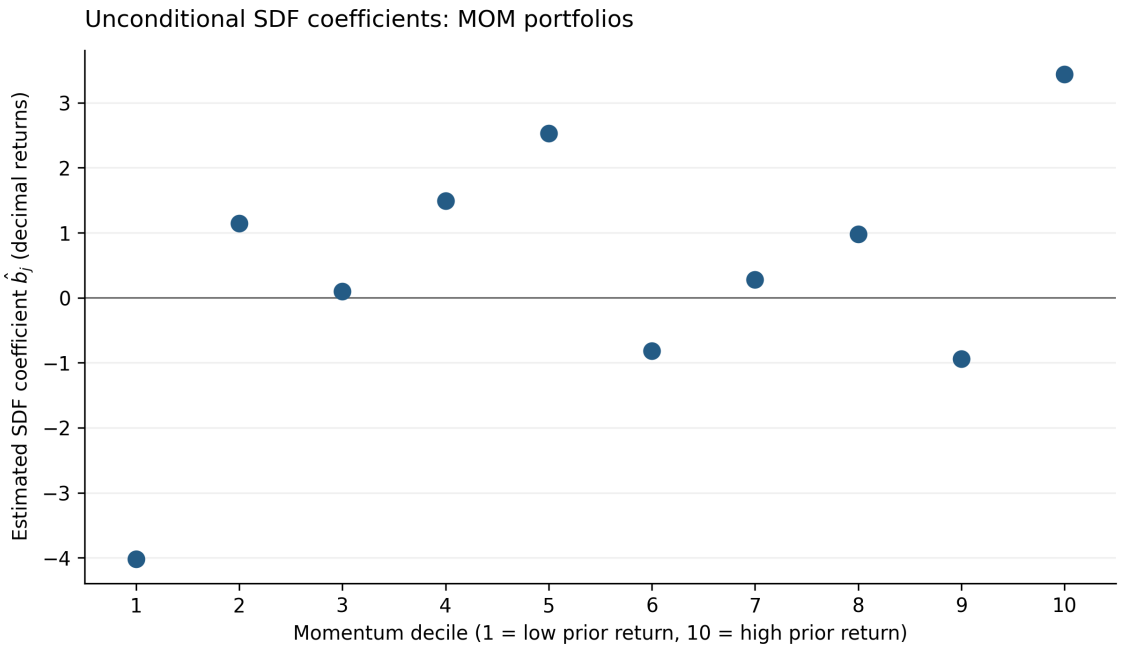


Figure 2: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$ for the ten momentum portfolios, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor T, as in part (a).

(c)

Following the same approach as in parts (a) and (b), I construct the unconditional linear SDF using the combined 20 BM and MOM portfolios. Table 3 reports the estimated SDF coefficients.

Figure 3 shows that the estimated coefficients vary considerably across both BM and MOM deciles, with positive and negative values and no clear monotonic pattern. Compared with parts (a) and (b), the coefficients change when the two sets of portfolios are combined, reflecting the additional covariance relationships and diversification opportunities.

Table 4 reports monthly Sharpe ratios of 0.1919, 0.2581, and 0.3045 for the BM, MOM, and combined maximum Sharpe ratio portfolios, respectively. The combined portfolio achieves the highest Sharpe ratio, consistent with the theoretical prediction that expanding the investment opportunity set cannot reduce the maximum attainable Sharpe ratio.

This result is also consistent with the SDF volatility bound discussed in class. Under the normalization $E[M^*] = 1$, the volatility of the constructed SDF equals the maximum Sharpe ratio. Thus, including both BM and MOM portfolios increases the maximum attainable Sharpe ratio and the volatility of the corresponding SDF.

Table 3: Combined SDF intercept and coefficients, rounded to four decimal places. The first ten coefficients correspond to BM portfolios and the next ten to MOM portfolios. Estimation uses decimal excess returns.

SDF intercept $\hat{a}$										1.0927
Decile j	1	2	3	4	5	6	7	8	9	10
BM	-1.9664	2.4739	0.5781	-3.7439	1.0655	2.6840	-4.6456	2.4707	3.7398	-0.2415
MOM	-4.4169	0.6354	-0.1712	1.9480	1.4236	-0.8781	-0.1639	0.9802	-1.2708	3.4473

Table 4: Monthly Sharpe ratios of $xR_t^* = \hat{b}' xR_t$, using the same 1,190 months from July 1927 through August 2026. Variances use divisor T.

Portfolio set	Monthly Sharpe ratio
BM	0.1919
MOM	0.2581
Combined	0.3045

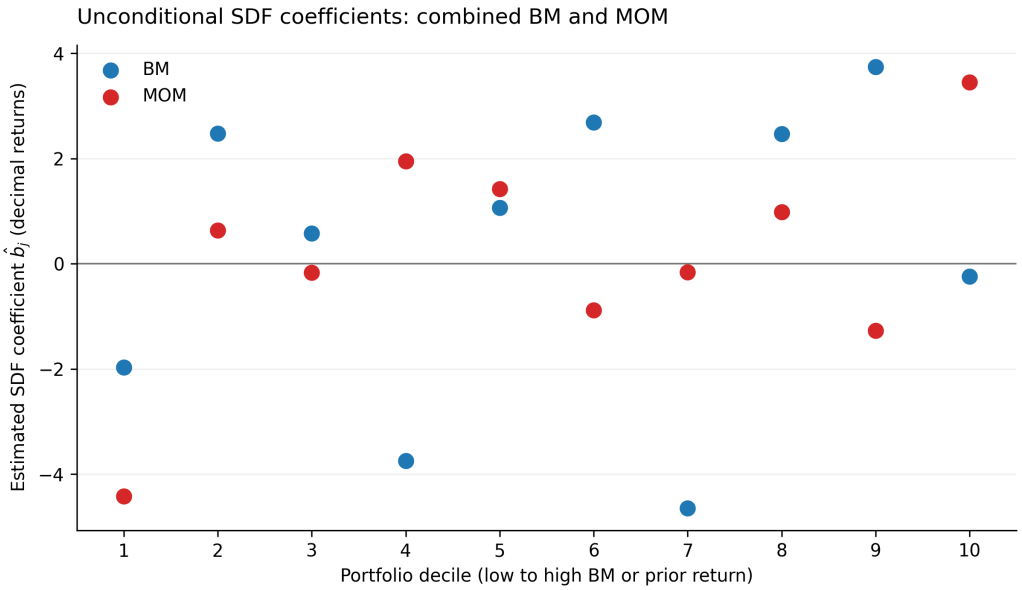


Figure 3: Coefficients in the combined BM–MOM SDF, normalized so that $\widehat{E}[\hat{M}^*] = 1$. Blue dots denote BM portfolios and red dots denote MOM portfolios. Returns are in decimals; the covariance matrix uses divisor T.

(d)

Table 5 reports the pricing errors of the 20 BM and MOM portfolios under the three SDFs constructed in parts (a), (b), and (c), using

$$\hat{\alpha}_p = \hat{E}[xR_p] - \widehat{\mathrm{Cov}}(-\hat{M}^*, xR_p).$$

The pricing errors are essentially zero for the portfolios used to construct each SDF. Specifically, the BM SDF prices all 10 BM portfolios, the MOM SDF prices all 10 MOM portfolios, and the combined SDF prices all 20 portfolios, with errors close to numerical precision.

However, the BM and MOM SDFs exhibit nonzero pricing errors when applied to portfolios outside their respective construction sets. The maximum absolute pricing error is 0.010157 for MOM portfolios under the BM SDF, compared with 0.003450 for BM portfolios under the MOM SDF.

These results suggest that an SDF constructed from one set of portfolios does not necessarily price another set of portfolios accurately. The combined SDF eliminates the in-sample pricing errors by incorporating both sets of portfolios. However, this exact in-sample fit does not necessarily imply good out-of-sample pricing performance.

Table 5: Monthly pricing errors $\hat{\alpha}_p$ under the BM, MOM, and combined SDFs, in decimal return units and rounded to six decimal places. The common sample is July 1927 through August 2026 (1,190 months); covariances use divisor T.

Portfolio	BM SDF	MOM SDF	Combined SDF
BM: Lo 10	0.000000	-0.001511	0.000000
BM: 2-Dec	0.000000	-0.000101	0.000000
BM: 3-Dec	0.000000	-0.000223	0.000000
BM: 4-Dec	0.000000	-0.000778	0.000000
BM: 5-Dec	0.000000	0.000504	0.000000
BM: 6-Dec	0.000000	0.001261	0.000000
BM: 7-Dec	0.000000	0.000281	0.000000
BM: 8-Dec	0.000000	0.002376	0.000000
BM: 9-Dec	0.000000	0.003450	0.000000
BM: Hi 10	0.000000	0.003198	0.000000
MOM: Lo PRIOR	-0.010157	0.000000	0.000000
MOM: PRIOR 2	-0.005051	0.000000	0.000000
MOM: PRIOR 3	-0.003272	0.000000	0.000000
MOM: PRIOR 4	-0.001488	0.000000	0.000000
MOM: PRIOR 5	-0.000882	0.000000	0.000000
MOM: PRIOR 6	-0.000263	0.000000	0.000000
MOM: PRIOR 7	0.000509	0.000000	0.000000
MOM: PRIOR 8	0.001902	0.000000	0.000000
MOM: PRIOR 9	0.002240	0.000000	0.000000
MOM: Hi PRIOR	0.005418	0.000000	0.000000

(e)

Following Question 2(c), I re-estimate the SDF using monthly excess returns of the 20 BM and MOM decile portfolios from July 1927 through December 1989:

$$\hat{M}_t^* = \hat{a} - \hat{\mathbf{b}}' \mathbf{x}\mathbf{R}_t.$$

This gives the estimated intercept $\hat{a}$ and coefficient vector $\hat{\mathbf{b}}$.

I then hold the estimated coefficients fixed and construct the out-of-sample SDF using monthly excess returns from January 1990 through August 2026:

$$\hat{M}_{OS,t}^* = \hat{a} - \hat{\mathbf{b}}' \mathbf{x}\mathbf{R}_t.$$

For each of the 20 portfolios, I calculate the out-of-sample pricing error as

$$\hat{\alpha}_{OS,p} = \widehat{E}[xR_p] - \widehat{\mathrm{Cov}}(-\hat{M}_{OS}^*, xR_p),$$

where both the expectation and covariance are estimated using only the out-of-sample observations. The resulting pricing errors are reported in Table 6. All 20 portfolios exhibit positive out-of-sample pricing errors, suggesting that the estimated SDF does not fully capture their average excess returns.

Table 6: Out-of-sample monthly pricing errors $\hat{\alpha}_{OS,p}$, in decimal return units and rounded to six decimal places. The SDF is estimated using July 1927 through December 1989 (750 months), with its intercept and coefficients held fixed for evaluation over January 1990 through August 2026 (440 months). Covariances use divisor T_{OS}.

Portfolio	Out-of-sample pricing error
BM: Lo 10	0.010971
BM: 2-Dec	0.007858
BM: 3-Dec	0.009002
BM: 4-Dec	0.006957
BM: 5-Dec	0.007986
BM: 6-Dec	0.007178
BM: 7-Dec	0.006977
BM: 8-Dec	0.005923
BM: 9-Dec	0.008269
BM: Hi 10	0.009709
MOM: Lo PRIOR	0.027055
MOM: PRIOR 2	0.016483
MOM: PRIOR 3	0.013905
MOM: PRIOR 4	0.012077
MOM: PRIOR 5	0.009180
MOM: PRIOR 6	0.008297
MOM: PRIOR 7	0.007601
MOM: PRIOR 8	0.006444
MOM: PRIOR 9	0.004322
MOM: Hi PRIOR	0.005453

Question 3

(a)

The investor solves

$$\max_{\varpi_t} E_t[U(W_{t+1})] \quad \text{s.t.} \quad W_{t+1} = W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big).$$

Assume that $W_t > 0$, $W_{t+1} > 0$ almost surely, and that the optimal portfolio choice is interior. Under the usual regularity conditions, we can derive the first-order condition by differentiating expected utility w.r.t. ϖ_t.

This problem can be written as

$$\max_{\varpi_t} E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big].$$

$$\frac{\partial E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big]}{\partial \varpi_t} = U'(W_{t+1}) \cdot W_t \cdot x R_{t+1}.$$

$$U(w) = \frac{w^{1-\gamma}}{1-\gamma}, \qquad U'(w) = w^{-\gamma}.$$

Thus, $U'(W_{t+1}) = W_{t+1}^{-\gamma}$.

$$\begin{aligned}\frac{\partial E_t\big[U\big(W_t\big(R_{f,t+1} + \varpi_t' x R_{t+1}\big)\big)\big]}{\partial \varpi_t} &= E_t\big[U'(W_{t+1}) \cdot W_t\big] \\ &= E_t\Big[W_{t+1}^{-\gamma} \cdot W_t \cdot x R_{t+1}\Big] = 0.\end{aligned}$$

$$E_t\left[\frac{W_t}{W_{t+1}^{\gamma}} x R_{t+1}\right] = W_t E_t\left[\frac{1}{W_{t+1}^{\gamma}} x R_{t+1}\right] = 0.$$

Since $W_{t+1}^{\gamma} = (W_t \cdot R_{w,t+1})^{\gamma}$,

$$E_t\left[\frac{1}{(W_t \cdot R_{w,t+1})^{\gamma}} x R_{t+1}\right] = W_t^{-\gamma} E_t\Big[R_{w,t+1}^{-\gamma} x R_{t+1}\Big] = 0.$$

Since $W_t > 0$, we have

$$E_t\Big[R_{w,t+1}^{-\gamma} x R_{t+1}\Big] = 0.$$

(b)

Table 7: GMM estimates using the supplied equity/risk-free log returns, December 1927 through December 2021. Values are rounded to four decimal places.

Statistic	Value
Risk aversion $\hat{\gamma}$	1.9061
Standard error $SE(\hat{\gamma})$	0.6026
Jacobian $\hat{G}$	-0.0480
Long-run variance $\hat{S}$	0.9439
Sample moment $\bar{g}_T(\hat{\gamma})$	0.0000
Observations T	1129
HAC lag L	26